



UPSC Prelims

Science & Tech

Class 3

Biotechnology

Common Doubts



- Enzymes for protein synthesis vs enzymes for energy creation
- DNA 'change' after viral integration
- Why RNA and why single stranded
- Proteins eaten vs Proteins manufactured
- Other viruses and How they work

Basics of Biotechnology



- DNA to RNA : Transcription
- RNA to protein : Translation
- Creation of mRNA
- Gene vs Genome
- DNA fingerprinting

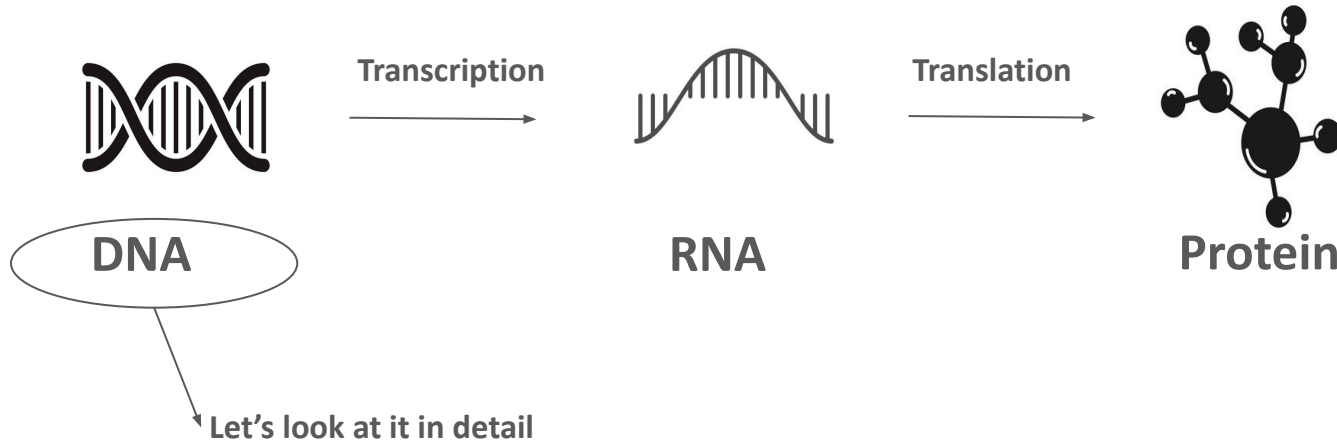


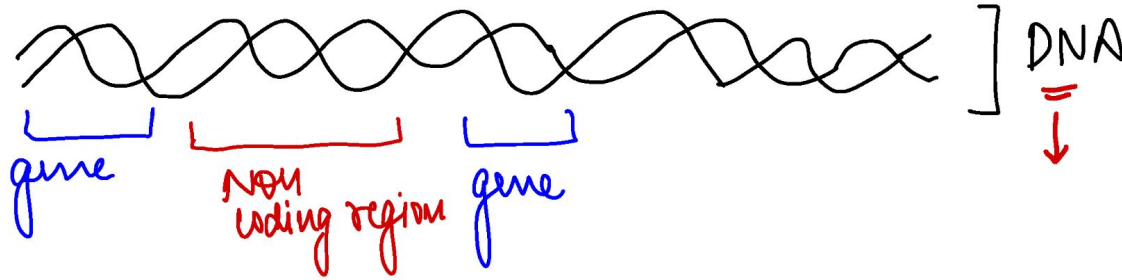
- Nucleic acid is the genetic material of all organisms without any exception.
- Configuration of nuclear material : number or strands / type : DNA or RNA varies.

Basics of Biotechnology



- DNA to RNA : Transcription
- RNA to protein : Translation
- Creation of mRNA
- Gene vs Genome
- DNA fingerprinting





Not all parts of DNA → code for a protein
coding part → genes.
→ code for RNA → code for protein

Question

Between two humans, what will be more similar ?

Genes (Coding Region)

Non Coding Region

Non coding Region

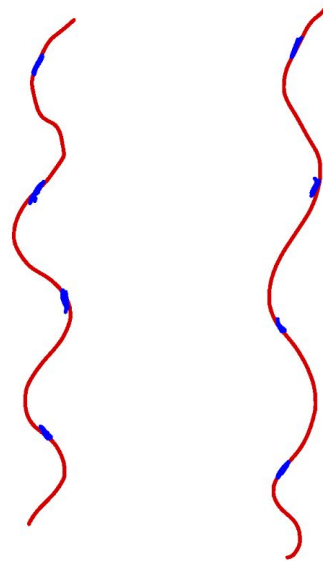
Many differences in between humans

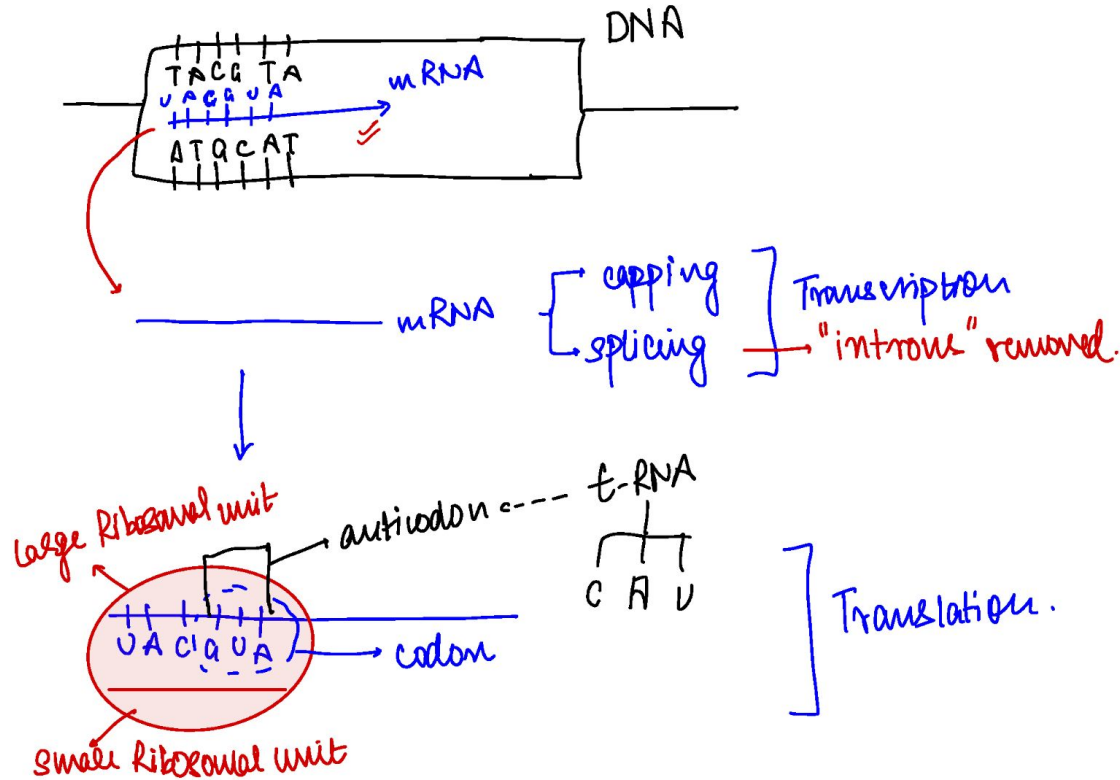


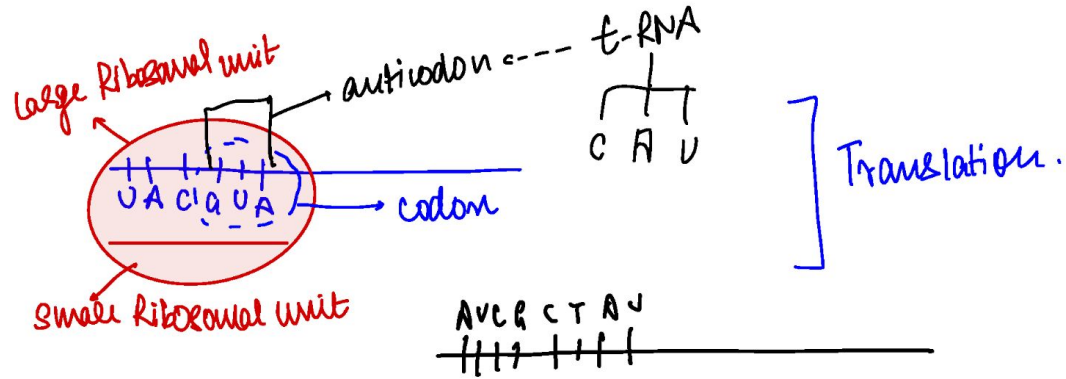
application



DNA fingerprinting



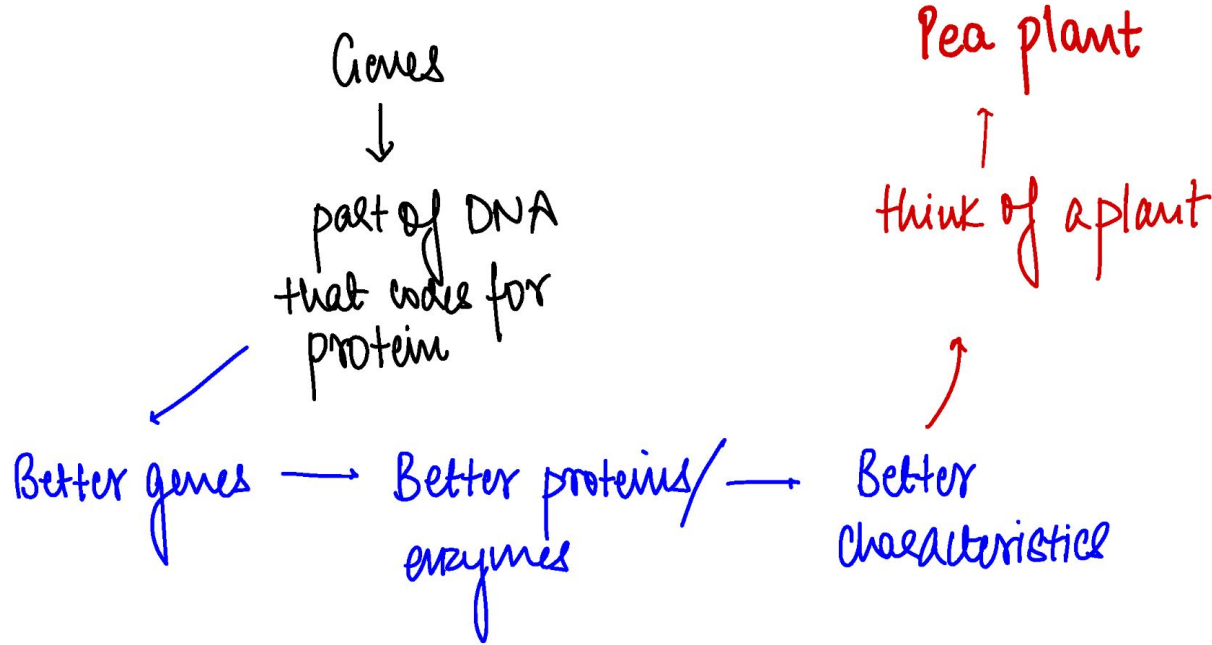


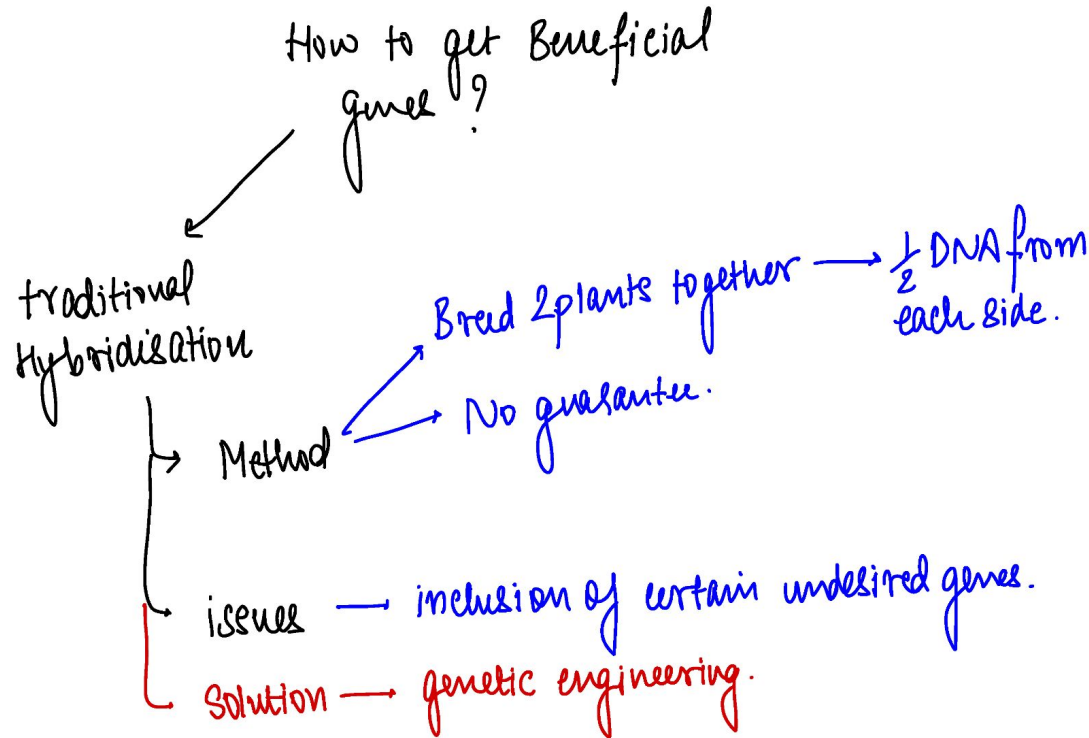


every 3 nucleotides (A, U, C, T) $\rightarrow$ code for an amino acid

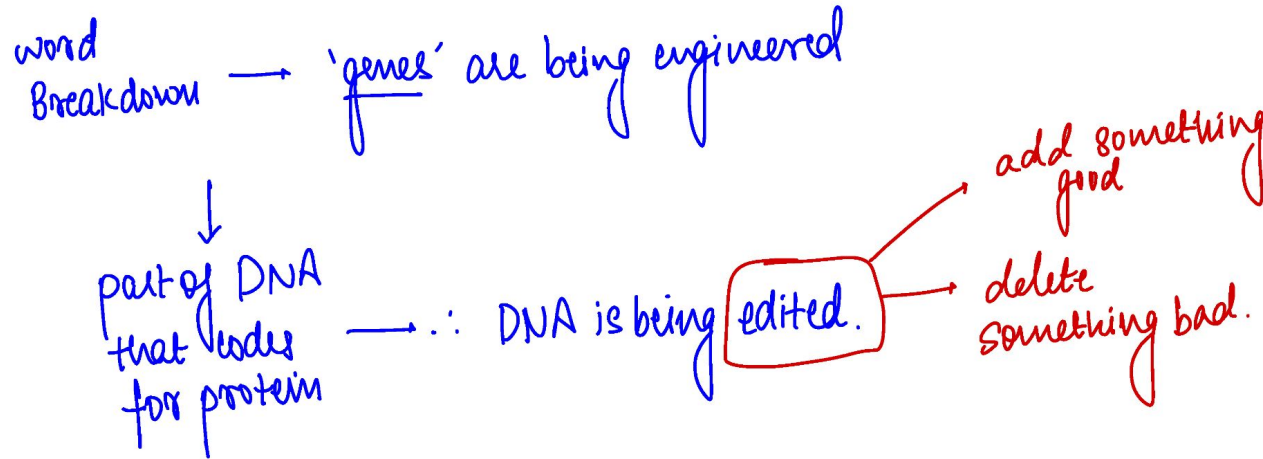
amino acids $\rightarrow$ Make up proteins

These amino acids are brought by 't' $\rightarrow$ transfer RNA.

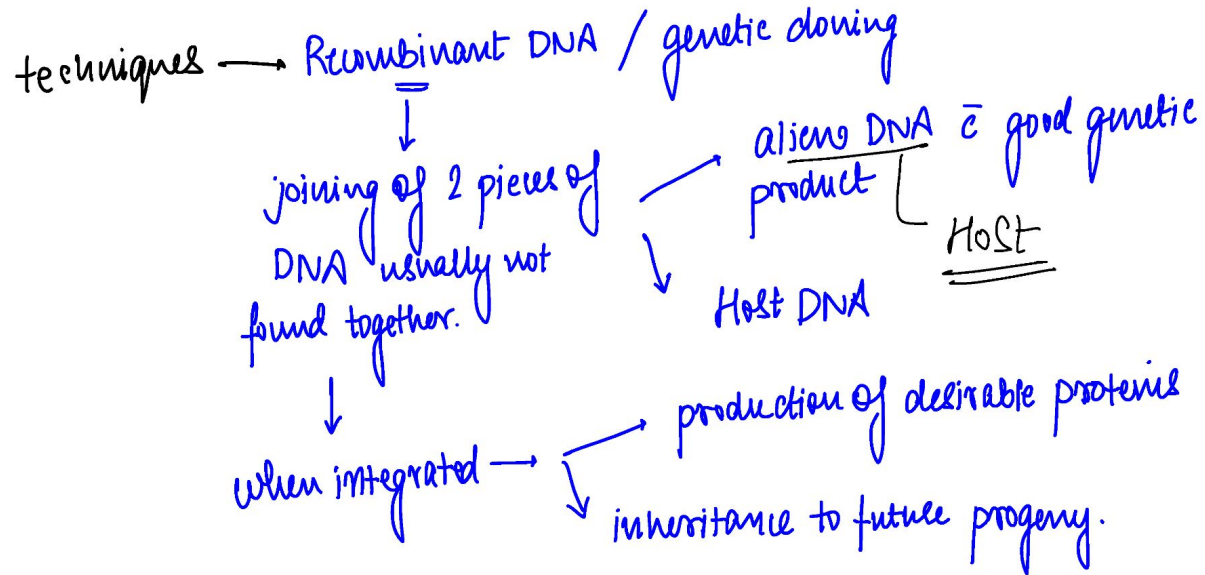




Genetic engineering



Genetic Engineering



Biotechnology : Introduction



- Biotechnology deals with techniques of using live organisms or enzymes from organisms to produce products and processes useful to humans.
- Used in a restricted sense today, to refer to such of those processes which use genetically modified organisms to achieve the same on a larger scale.
- European Federation of Biotechnology (EFB) definition is as follows : The integration of natural science and organisms, cells, parts thereof, and molecular analogues for products and services.

2 principles of Biotechnology



Genetic engineering : Techniques to alter the chemistry of genetic material (DNA and RNA), to introduce these into host organisms and thus change the phenotype of the host organism.

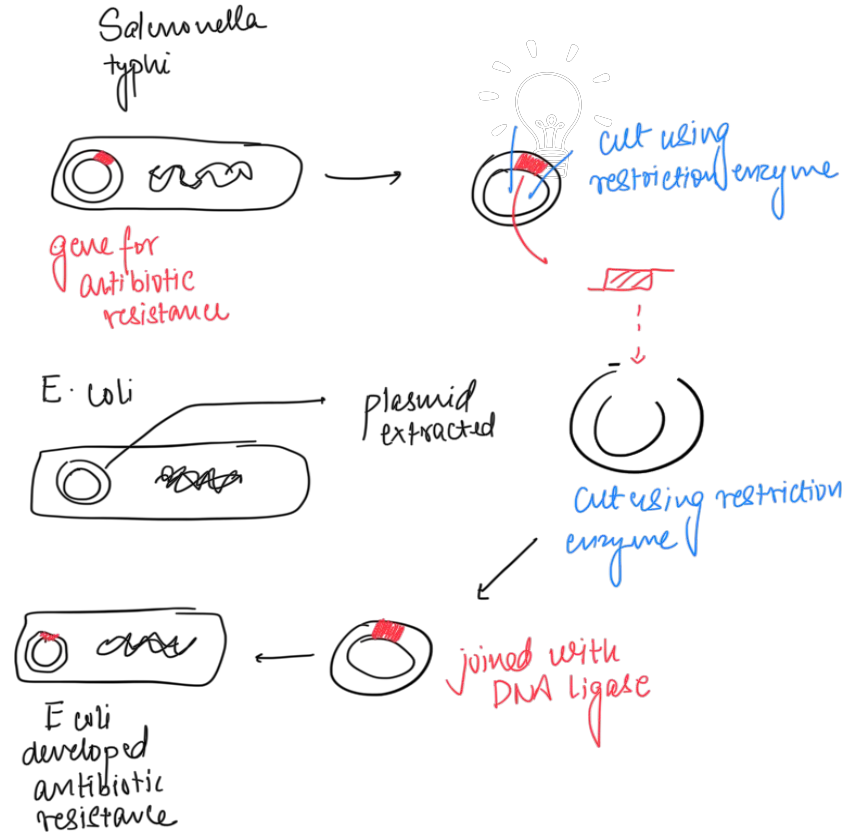
Bioprocess engineering : Maintenance of sterile (microbial contamination-free) ambience in chemical engineering processes to enable growth of only the desired microbe/eukaryotic cell in large quantities for the manufacture of biotechnological products like antibiotics, vaccines, enzymes, etc.

Recombinant DNA

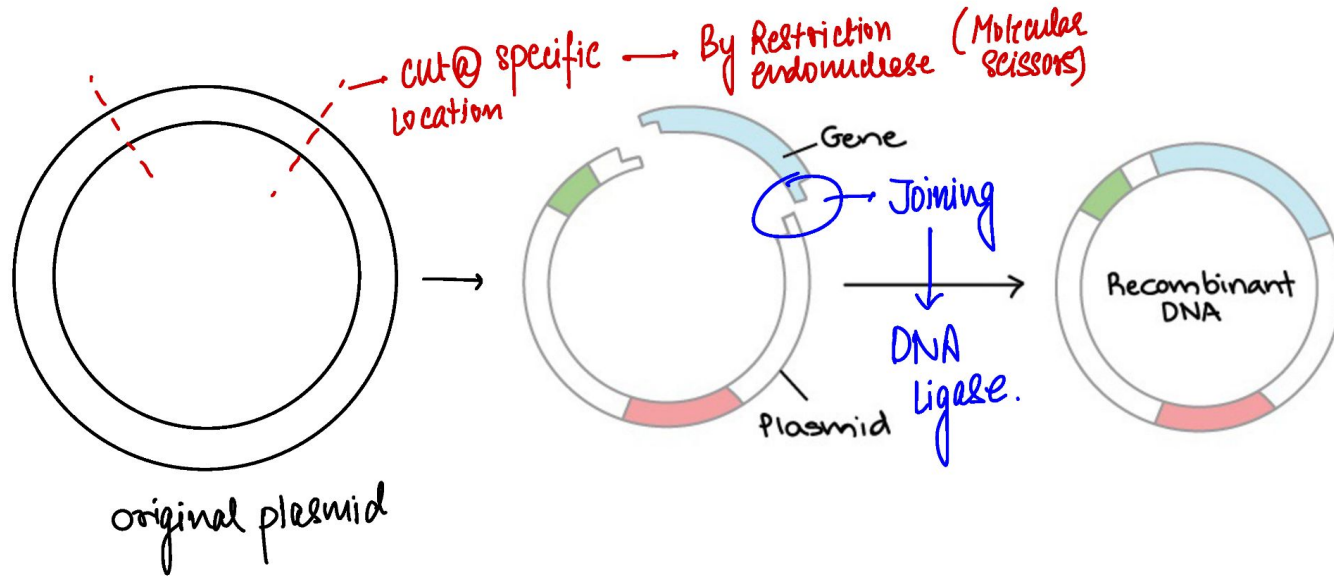


- first instance of the construction of an artificial recombinant DNA molecule : linking a gene encoding antibiotic resistance with a native **plasmid** (autonomously replicating circular extra-chromosomal DNA) of *Salmonella typhimurium* by Stanley Cohen and Herbert Boyer in 1972.

Cohen and Boyer experiment



Plasmid → autonomously replicating circular extra chromosomal DNA
acts as a vector. → Helps in Integration.

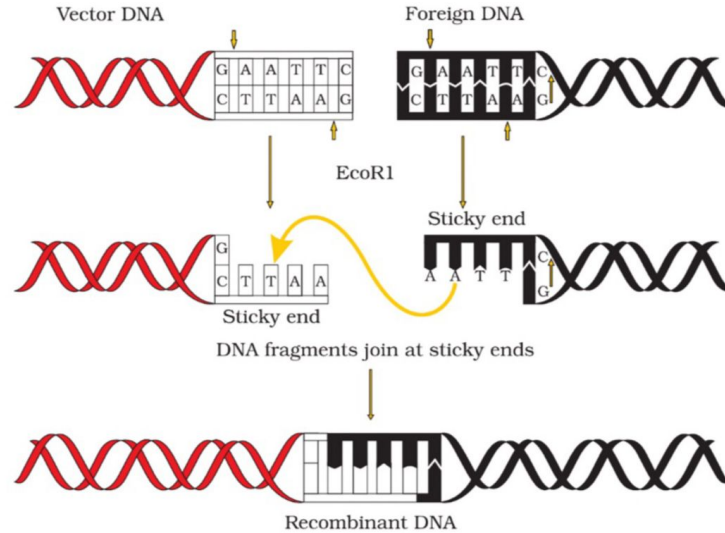


Action of Restriction enzyme

Scissor → cut

The enzyme cuts both DNA strands at the same site

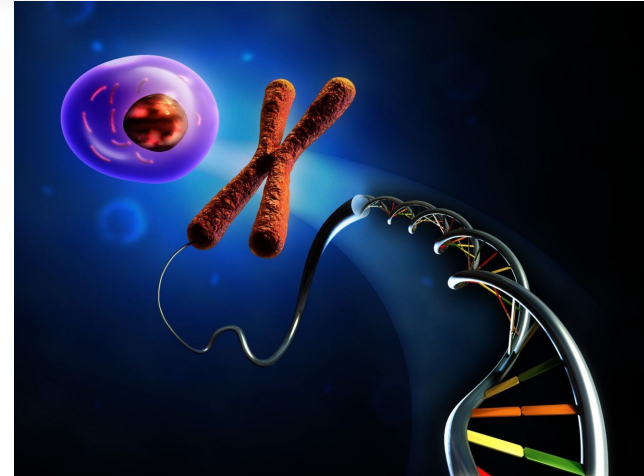
EcoRI cuts the DNA between bases G and A only when the sequence GAATTC is present in the DNA



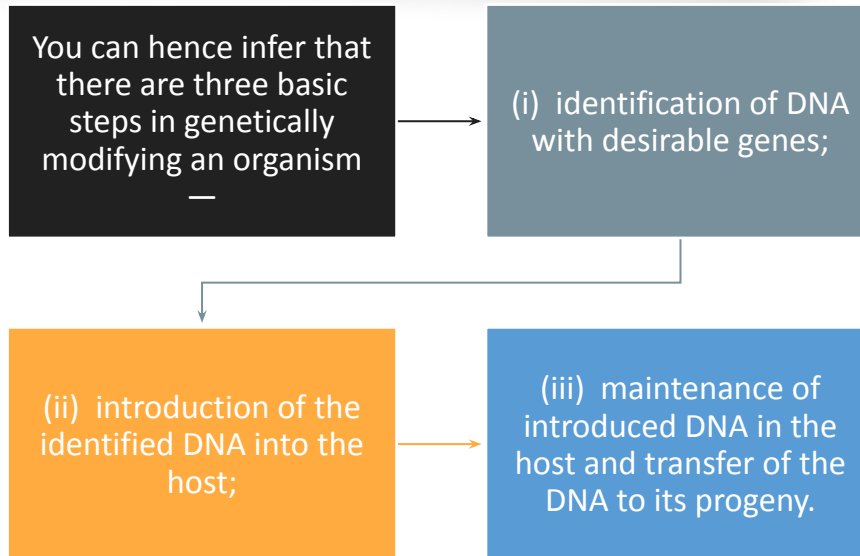
Recombinant DNA technology



- A plasmid can be used as vector to deliver an alien piece of DNA into the host organism.
- The linking of antibiotic resistance gene with the plasmid vector became possible with the enzyme DNA ligase
- This makes a new combination of circular autonomously replicating DNA created *in vitro* and is known as recombinant DNA.
- it could replicate using the new host's DNA polymerase enzyme and make multiple copies.



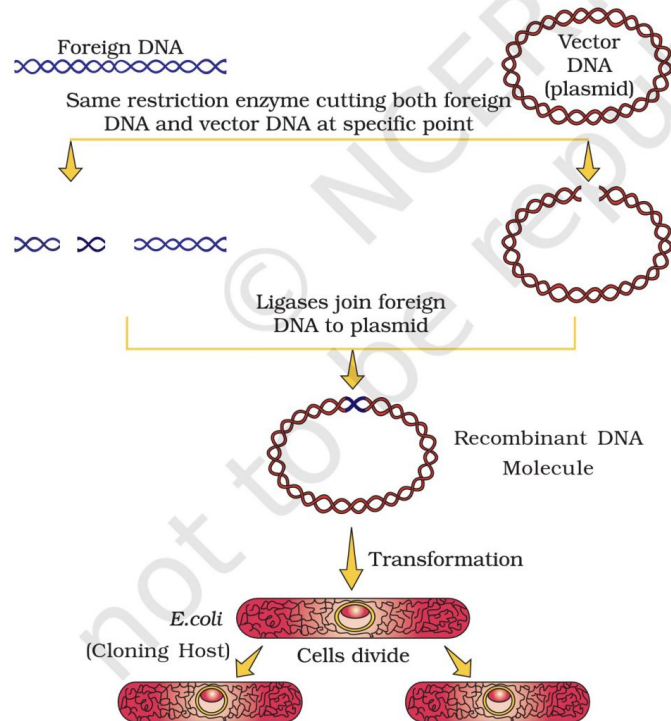
Genetic Modification



Tools of RDT : Restriction Enzymes



- The first restriction endonuclease was Hind II
- It was found that Hind II always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs : this specific sequence was called Recognition Sequence
- Restriction enzymes belong to a larger class of enzymes called nucleases. These are of two kinds; exonucleases and endonucleases.
- Exonucleases remove nucleotides from the ends of the DNA whereas, endonucleases make cuts at specific positions within the DNA.
- Each restriction endonuclease recognises a specific palindromic nucleotide sequences in the DNA.
- When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of 'sticky-ends' and, these can be joined together (end-to-end) using DNA ligases
- Separation and isolation of DNA fragments : Use of Agarose gel electrophoresis



Need of Vectors



- The likely fate of a piece of DNA, which is somehow transferred into an alien organism would not be able to multiply itself in the progeny cells of the organism.
- when it gets integrated into the genome of the recipient, it may multiply and be inherited along with the host DNA.
- This is because the alien piece of DNA has become part of a chromosome, which has the ability to replicate.



Tool of RDT : Cloning Vectors



- Plasmids and Bacteriophages
- Bacteriophages because of their high number per cell, have very high copy numbers of their genome within the bacterial cells.
- If we are able to link an alien piece of DNA with bacteriophage or plasmid DNA, we can multiply its numbers equal to the copy number of the plasmid or bacteriophage.

Steps of RDT

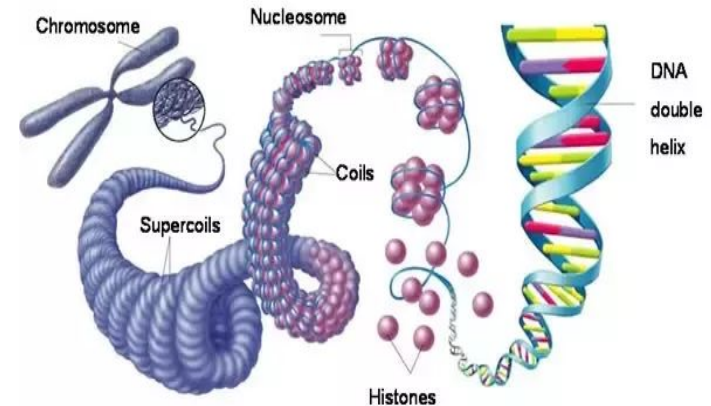


1. Isolation of DNA : pure form
2. Cutting at specific places : RE + gel electrophoresis
3. Polymerase Chain Reaction : amplification of gene of interest
4. Ligation
5. Insertion into host : selectable marker
6. Recombinant Protein production
7. Downstream Processing

Isolation of DNA fragments



1. We need pure form of DNA : break open the cell and release DNA using certain enzymes.
2. This can be achieved by treating
3. the bacterial cells/plant or animal tissue with enzymes such as lysozyme (bacteria), cellulase (plant cells), chitinase (fungus).
4. You do not need to remember these enzymes.
5. Genes are located on DNA that is wound into chromosomes using histones.



Cutting at specific places : RE + gel electrophoresis



1. Cut the source DNA and vector DNA with the same enzyme
2. Agarose gel electrophoresis : separates the DNA fragments by size.
3. This checks progression of restriction enzyme work

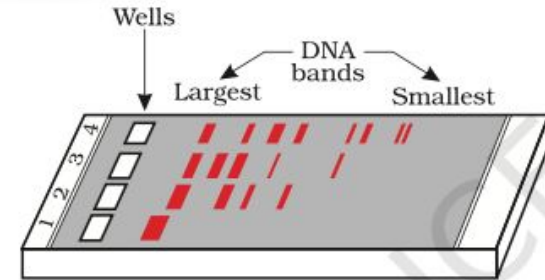
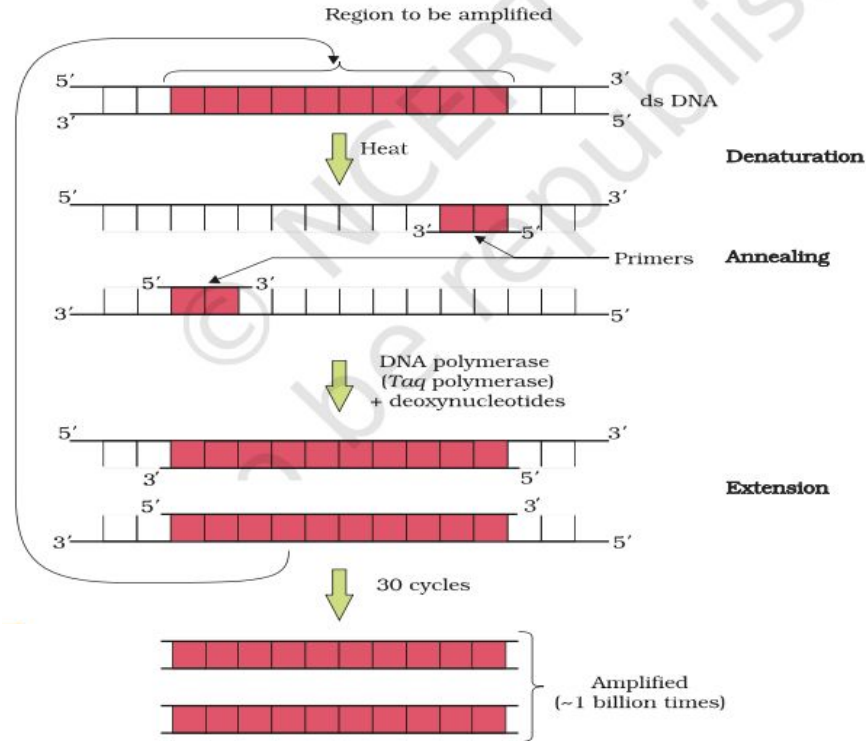
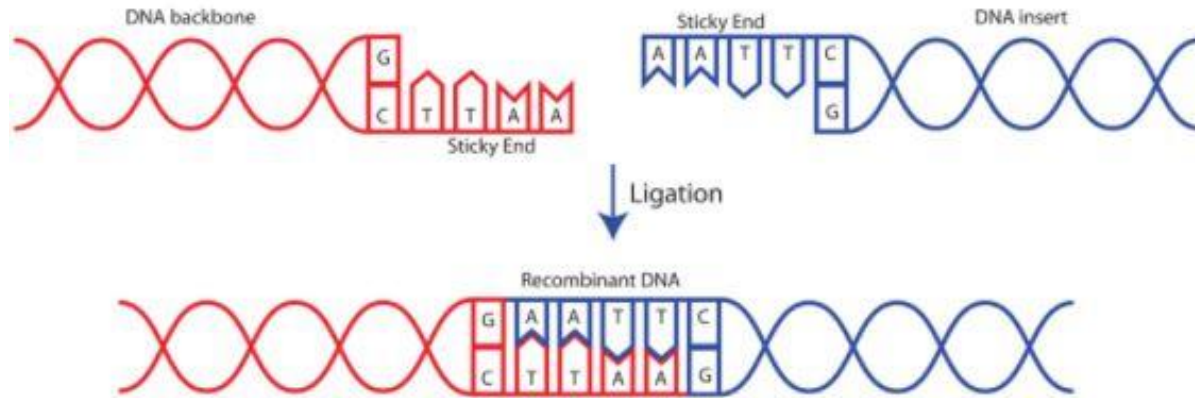


Figure 11.3 A typical agarose gel electrophoresis showing migration of undigested (lane 1) and digested set of DNA fragments (lane 2 to 4)

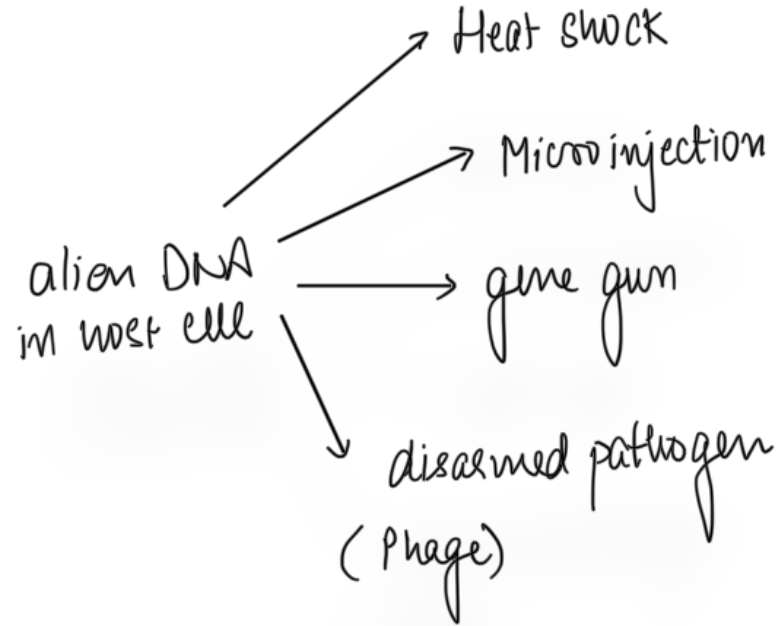
Polymerase chain reaction



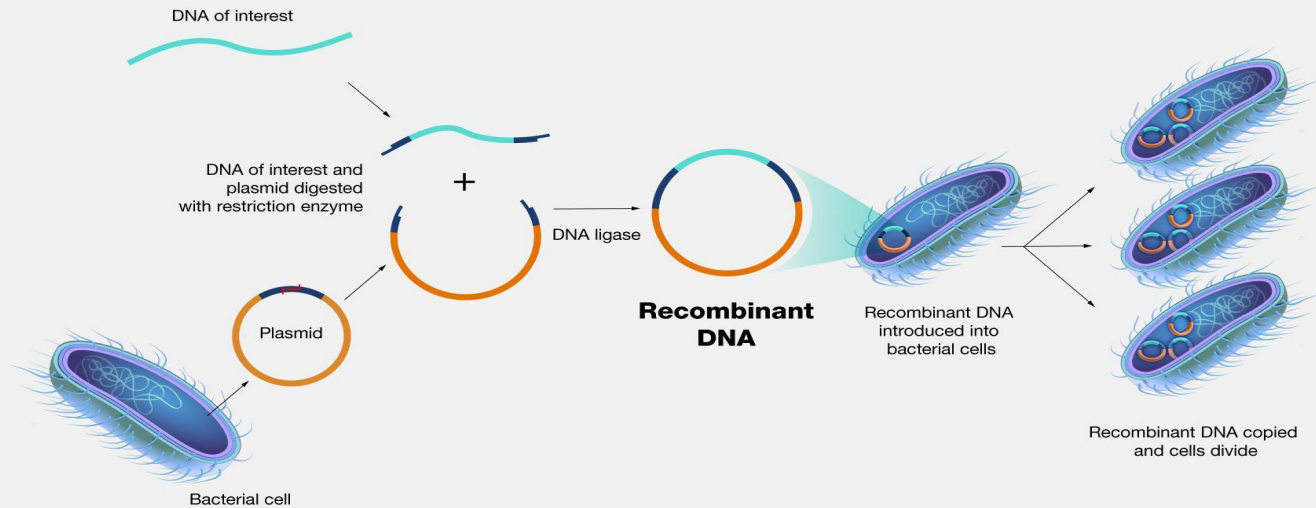
Ligation



Insertion into Host and
production of recombinant
protein



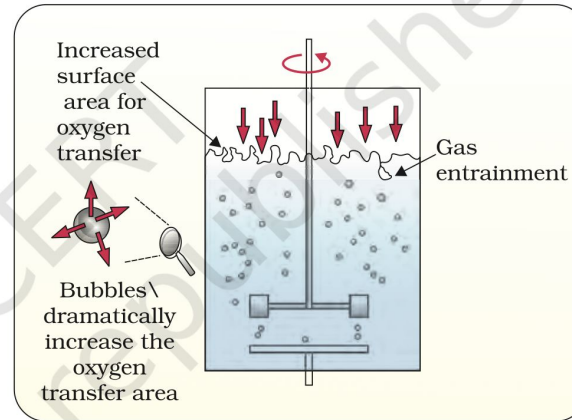
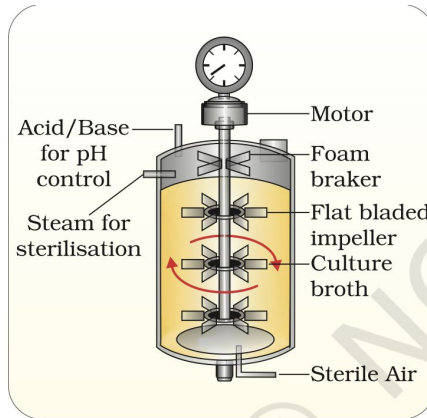
Production of Recombinant Protein



Obtaining foreign gene product



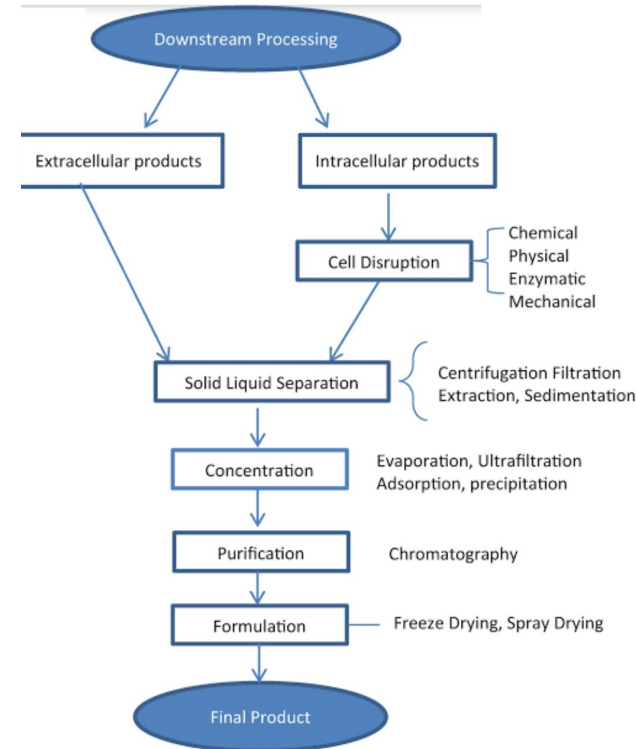
- Use of bioreactors to grow bacteria at a large scale

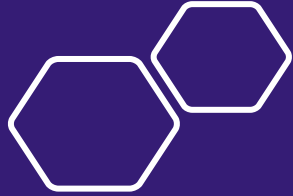


Downstream processing



- Use of bioreactors





Time for Doubts and Questions

5 minutes break

Take this time to revise what
you have learnt so far.

Applications of Biotechnology : Agriculture



- Question : what are the issues of Agrochemical based agriculture?



Solution : GM crops



Applications of genetic modification :

- (i) made crops more tolerant to abiotic stresses (cold, drought, salt, heat).
- (ii) reduced reliance on chemical pesticides (pest-resistant crops).
- (iii) helped to reduce post harvest losses.
- (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil).
- (v) enhanced nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice.
- (vi) Tailor made plants to deliver certain products : pharmaceuticals and fuels.

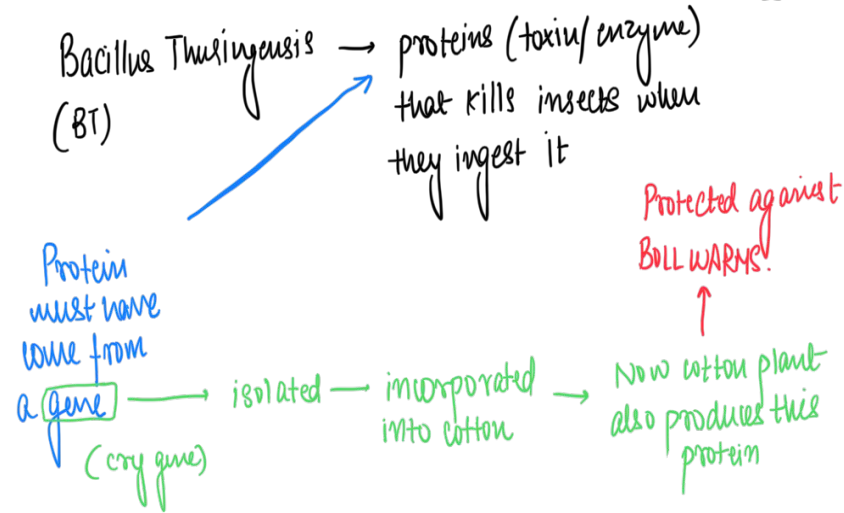
Ex : GM Algae for biofuels

Pest resistance in Detail



- Some strains of bacillus thuringensis produce proteins that kill certain insects (ex : bollworm). This protein is called BT toxin.
- Remains as protoxin in bacteria, but becomes ingested once ingested by a pest.
- Bt toxin gene has been cloned from the bacteria and been expressed in plants to provide resistance to insects without the need for insecticides.
- Specific bt toxin genes isolated from specific strains of bacteria and incorporated into plants. Choice of gene depends on crop and targeted pest.
- in effect this has created a bio-pesticide. Examples are Bt cotton, Bt corn, rice, tomato, potato and soyabean etc.
- Cry genes : CryIAC, cryIIAB : cotton bollworms
- CryIAB : corn borer

BT cotton and Bollworms



Method 2 : RNA interference method



RNAi takes place in all eukaryotic organisms as a method of cellular defense.

The main idea here is to silence the mRNA of the pathogen so that it cannot express its proteins.

This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA (silencing).

A nematode *Meloidogyne incognita* infects the roots of tobacco plants and causes a great reduction in yield.

Using *Agrobacterium* vectors, nematode-specific genes were introduced into the host plant. This allowed host (tobacco) to produce mRNA complementary to the mRNA of pathogen.

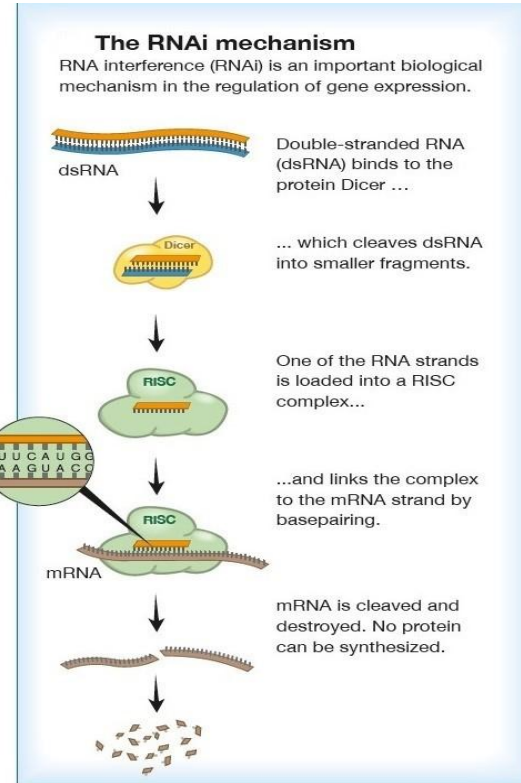
hence two RNA's being complementary to each other formed a double stranded (dsRNA) that initiated RNAi and thus, silenced the specific mRNA, of the nematode. Because of the formation of dsRNA due to binding of the 2 strands, the process of translation was stopped.

The consequence was that the parasite could not survive in a transgenic host expressing specific interfering RNA.

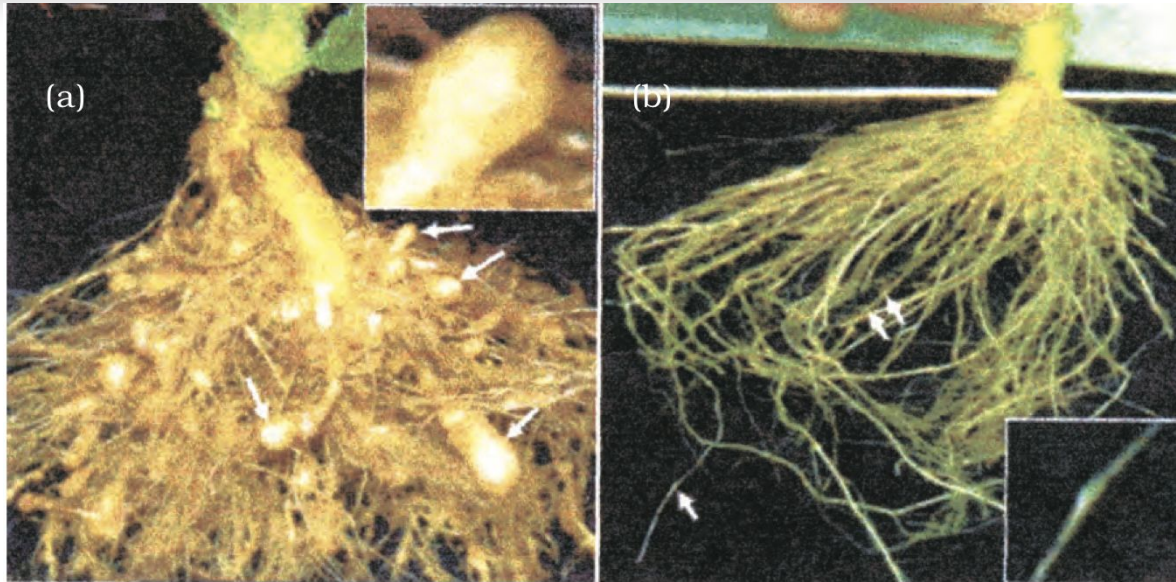
Rna Interference Mechanism



- Dicer protein
- RISC protein
- Formation of dsRNA due to introduction of nematode specific genes
- Stopping translation



RNA interference in tobacco : infection gone!



GM crops : use of mycorrhizza



- A mycorrhiza is a symbiotic relationship between a fungus and a root system.
- A network of fungus filaments surrounds the developing root or reaches the root cells directly.
- The hyphae possess a relatively broad surface area, which allows them to collect water and mineral ions from a significantly more amount of soil than the root.
- Mycorrhizal plants have tolerance to adverse soil, PH, high temp., drought and toxic heavy metals.
- It also helps plants resist certain diseases.
- The roots receive water and minerals from the fungus, and the mycorrhizae receive sugars and N-containing components from the roots.

GM Mustard : DMH 11



- It is a Genetically Modified Mustard
- DMH = Dhara Mustard Hybrid
- It is herbicide tolerant mustard
- Developed by the Centre for Genetic Manipulation of Crop Plants of Delhi University
- It was created by using “barnase/barstar” technology for genetic modification by adding genes from soil bacterium that makes mustard self-pollinating plant. It also allows for cross pollination and hybridisation.
- DMH -11 contains three genes viz. Bar gene, Barnase and Barstar sourced from soil bacterium.
- The bar gene had made plant resistant to herbicide named Basta.

GM Crops Challenges



1. Biosafety : elimination of wild species and pollution of gene pool
2. Safety with respect to human consumption
3. Legal disputes with respect to trait fees : Monsanto developed bt cotton technology called ?? (reveal later). Now seed production companies have to pay royalty to Monsanto.
4. Terminator gene technology : when such plants are planted, they will produce crops only once. Their resultant crop's seeds will be sterile. Farmer forced to buy new seeds every time
5. Genetic Divide : small and large farmers

GM crops : Regulation in India



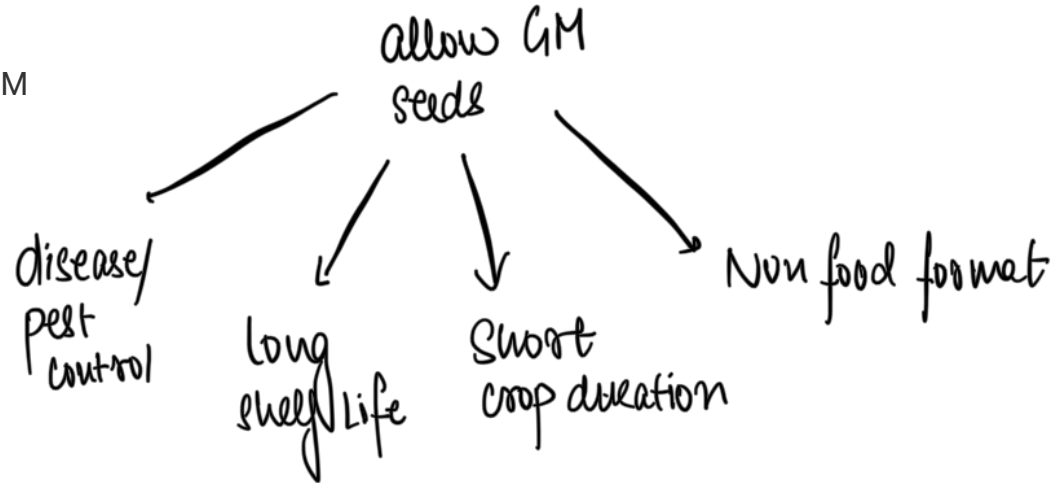
Under Environment Protection Act, 1986 :
GEAC (genetic engineering appraisal committee) conducts studies and recommends approving GM crop in India

Final permission : Ministry of Environment, Forests and Climate Change.

Economic Survey 2017 suggestions:



1. Allow GM seeds : low cost, without terminator genes
2. Encourage Domestic production of GM crops to end MNC monopoly



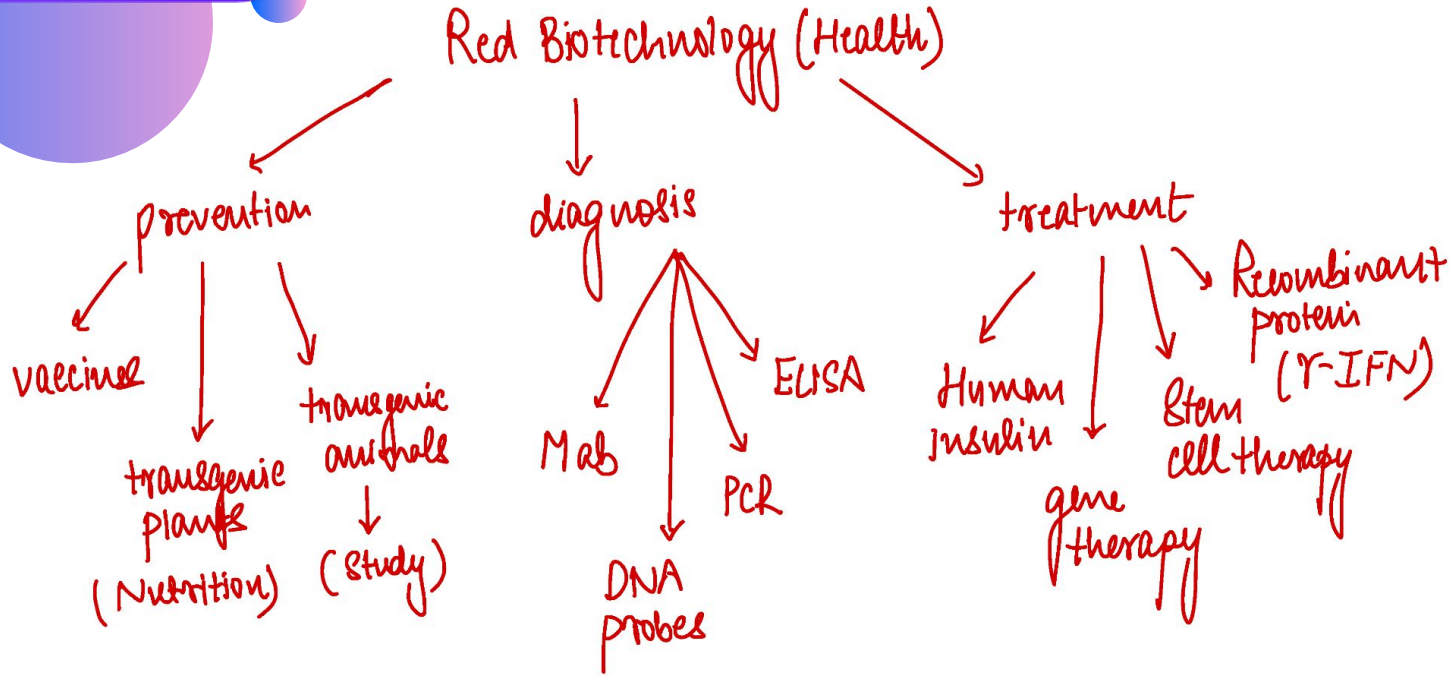
Applications of RDT/Biotechnology [India]

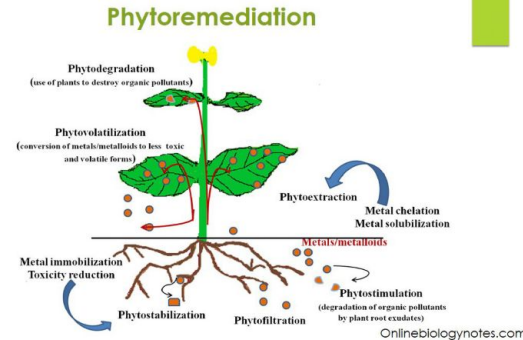
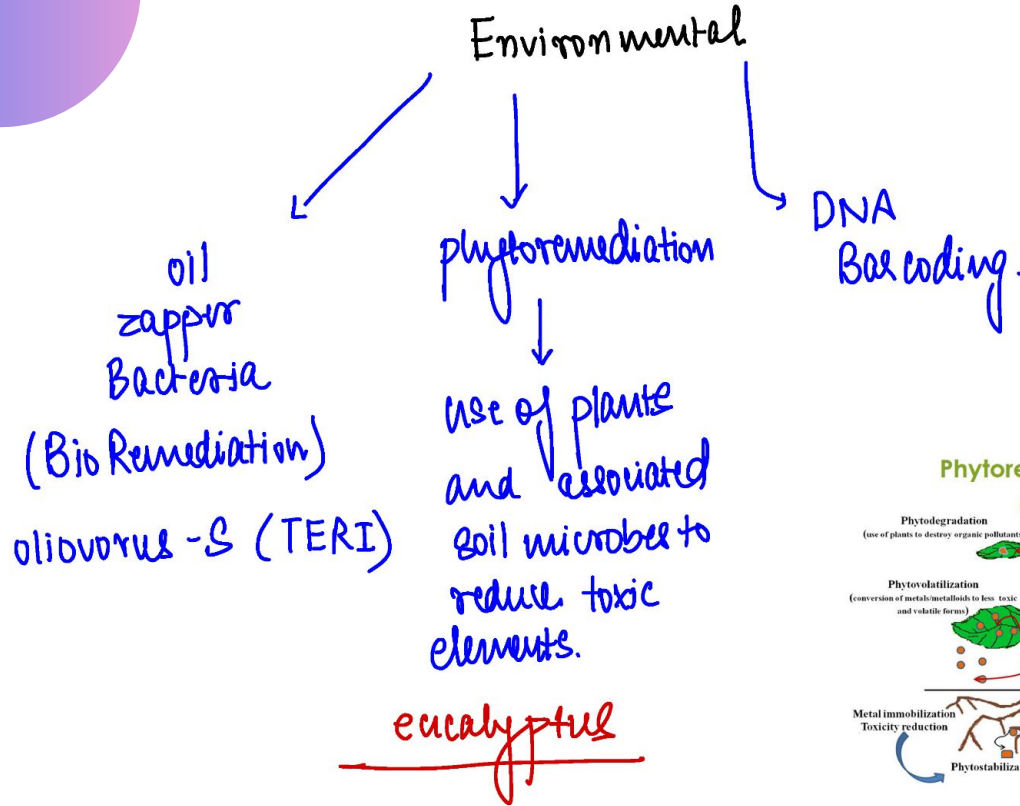
Green Biotech (Agriculture)

- (i) Yield
- (ii) Biopesticides: BT Brinjal
- (iii) HYV → GM Mustard
- (iv) Biofortification → golden Rice (vit-A)
- (v) Shelf life → Flavr Savr tomato (export)
- (vi) disease (Samba Masuri Rice)
- (vii) RNAi → plant vaccination
- (viii) cold tolerant strawberry
- (ix) Better Breed → cows, etc.

Blue Biotechnology (Marine)

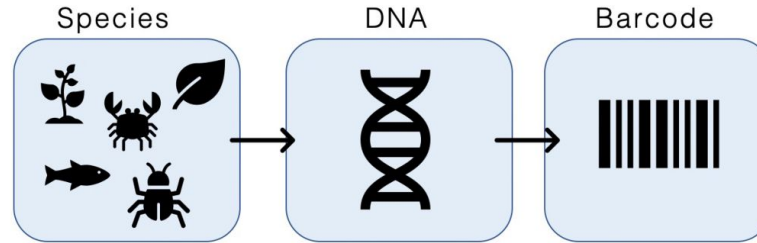
- aquaculture (sea-food cultivation)
- Transgenic fish → freeze resistant fish
- Conservation of species
- seaweed cultivation → ex: agar-agar.





DNA Barcoding

DNA barcoding is a method of species identification using a short section of DNA from a specific gene or genes. The premise of DNA barcoding is that by comparison with a reference library of such DNA sections (also called "sequences"), an individual sequence can be used to uniquely identify an organism to species, just as a supermarket scanner scans barcodes.

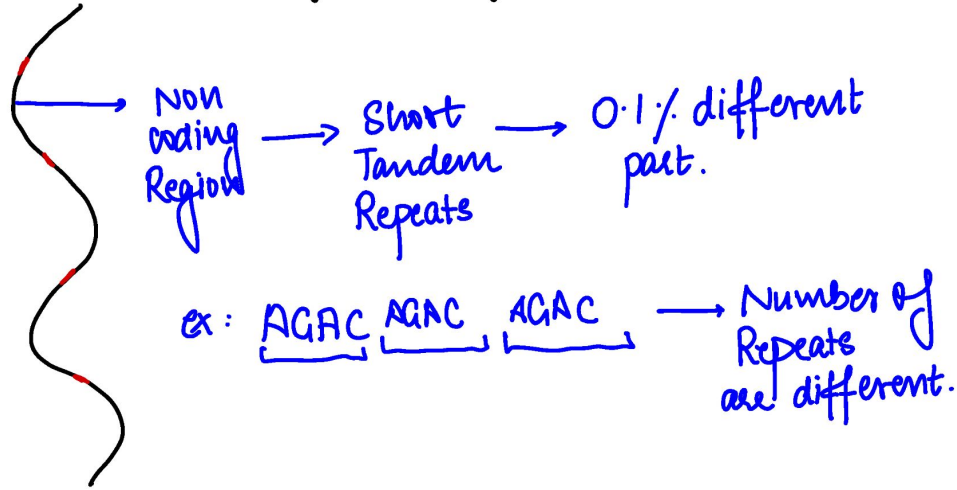


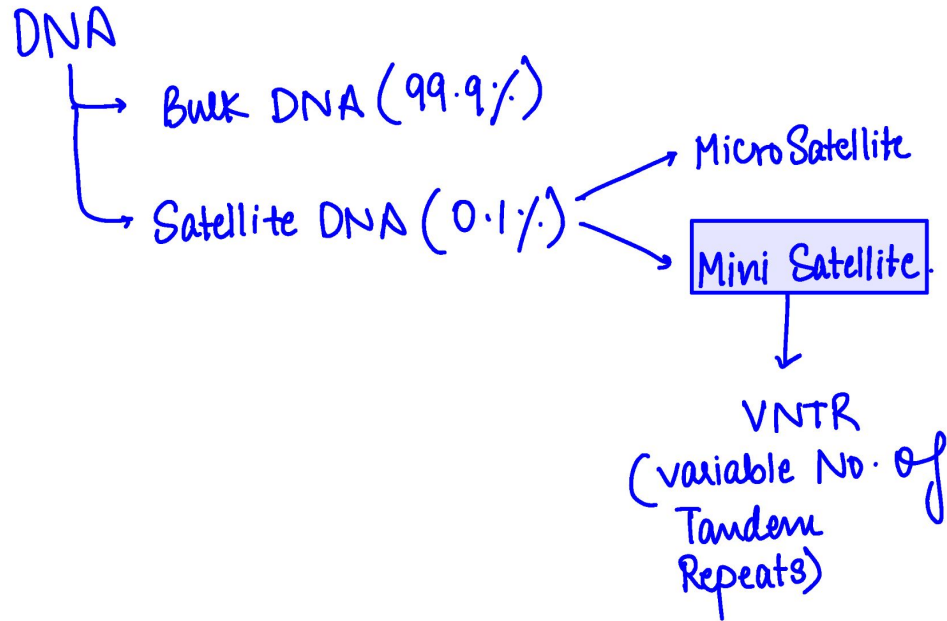
Forensic Biotechnology



- DNA fingerprinting : identification of criminal by use of biotechnology.
- Here we use something called variable number of tandem repeats (VNTR)
- Use to develop DNA bands in Gel electrophoresis and identify a person.
- Can be used in :
 - Criminal Investigation
 - Paternity testing
 - Immigration Cases
 - Child swapping cases
 - Disaster victim identification

DNA fingerprinting





Isolation and extraction



Digestion of DNA by endonuclease



Cut at specific place



Gel electrophoresis → But cant see.

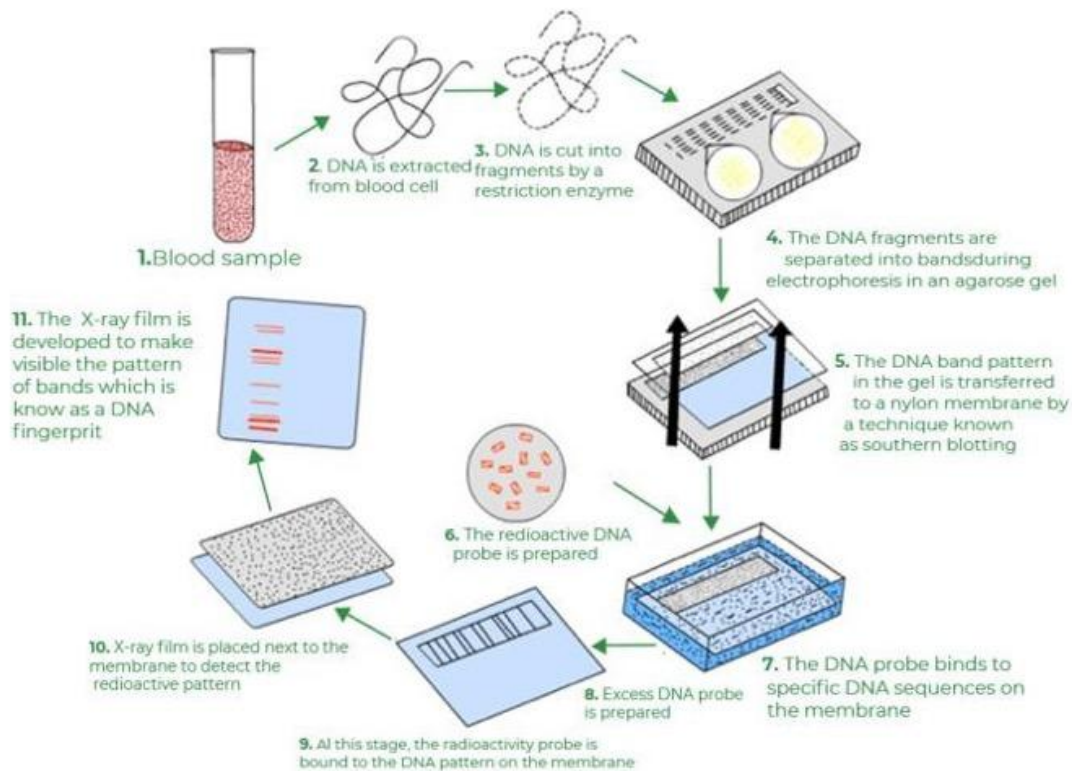


Southern Blotting → Nitrocellulose

autoradiography



Hybridisation
with Radioactive
probe



Test of 6 Questions!

**Treat this as a learning
exercise!**



Question 1

Q. With reference to the Genetically Modified mustard (GM mustard) developed in India, consider the following statements :

1. GM mustard has the genes of a soil bacterium that gives the plant the property of pest-resistance to a wide variety of pests.
2. GM mustard has the genes that allow the plant cross-pollination and hybridization.
3. GM mustard has been developed jointly by the IARI and Punjab Agricultural University.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Question 2

Q. 'RNA interference (RNAi)' technology has gained popularity in the last few years. Why?

1. It is used in developing gene silencing therapies.
2. It can be used in developing therapies for the treatment of cancer.
3. It can be used to develop hormone replacement therapies.
4. It can be used to produce crop plants that are resistant to viral pathogens.

Select the correct answer using the code given below:

- a) 1, 2 and 4
- b) 2 and 3
- c) 1 and 3
- d) 1 and 4 only

Question 3

Q. Recombinant DNA technology (Genetic Engineering) allows genes to be transferred

1. Across different species of plants.
2. From animals to plants.
3. From microorganisms to higher organisms.

Select the correct Solution using the codes given below.

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Question 4

Q. Mycorrhizal Biotechnology has been used in rehabilitating degraded sites because mycorrhiza enables the plants to -

1. Resist drought and increase absorptive area
2. Tolerate extremes of pH
3. Resist disease infestation

Select the correct answer using the codes given below:

- a) 1 only
- b) 1 and 2 only
- c) 1, 2 and 3 only
- d) 2 and 3 only

Question 5

Q. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future?

1. Genome sequencing can be used to identify genetic markers for disease resistance and drought tolerance in various crop plants.
2. This technique helps in reducing the time required to develop new varieties of crop plants.
3. It can be used to decipher the host-pathogen relationships in crops.

Question 6

Q. Bollgard I and Bollgard II technologies are mentioned in the context of:

- A. clonal propagation of crop plants
- B developing genetically modified crop plants
- C. production of plant growth substances
- D. production of biofertilizers

Congratulations on giving the test!

Let's Discuss The Answers.



Question 1 (Year 2018)

Q. With reference to the Genetically Modified mustard (GM mustard) developed in India, consider the following statements :

1. GM mustard has the genes of a soil bacterium that gives the plant the property of pest-resistance to a wide variety of pests.
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Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 only**
- c) 2 and 3 only
- d) 1, 2 and 3

Answer: B

- Dhara Mustard Hybrid-11, otherwise known as DMH - 11, is a genetically modified hybrid variety of the mustard species *Brassica juncea*.
- It was developed by Professor Deepak Pental from the University of Delhi, with the aim of reducing India's demand for edible oil imports. DMH - 11 was created through transgenic technology, primarily involving the Bar, Barnase and Barstar gene system.
- The Barnase gene confers male sterility, while the Barstar gene restores DMH - 11's ability to produce fertile seeds.
- The insertion of the third gene Bar enables DMH - 11 to produce phosphinothricin-Nacetyl-transferase, the enzyme responsible for Glufosinate resistance.
- This hybrid mustard variety has come under intense public scrutiny, mainly due to concerns regarding DMH - 11's potential to adversely affect the environment as well as consumer health.
- DMH - 11 was found not to pose any food allergy risks, and has demonstrated increased yields over existing mustard varieties. Conflicting details and results regarding the field trials and safety evaluations conducted on DMH - 11 have delayed its approval for commercial cropping

Question 2 (Year 2019)

Q. 'RNA interference (RNAi)' technology has gained popularity in the last few years. Why?

1. It is used in developing gene silencing therapies.
2. It can be used in developing therapies for the treatment of cancer.
3. It can be used to develop hormone replacement therapies.
4. It can be used to produce crop plants that are resistant to viral pathogens.

Select the correct answer using the code given below:

- a) 1, 2 and 4**
b) 2 and 3
c) 1 and 3
d) 1 and 4 only

Answer: A

- RNA interference (RNAi) or Post-Transcriptional Gene Silencing (PTGS) is a conserved biological response to double-stranded RNA that mediates resistance to both endogenous parasitic and exogenous pathogenic nucleic acids, and regulates the expression of protein-coding genes
- Although most research is currently looking into the applications of RNAi in cancer treatment, the list of possible applications is extensive.
- RNAi could potentially be used to treat viruses, bacterial diseases, parasites, maladaptive genetic mutations, control drug consumption, provide pain relief, and even modulate sleep
- RNA interference has been used for applications in biotechnology and is nearing commercialization in other fields.
- RNAi has resulted in the invention of novel crops such as nicotine-free tobacco, decaffeinated coffee, nutrient fortified vegetation, and hypoallergenic crops.
- The genetically-engineered Arctic apples received FDA approval in 2015.
- The apples were produced by RNAi suppression of the PPO (polyphenol oxidase) gene, making apple varieties that will not undergo browning after being sliced.
- PPO-silenced apples are unable to convert chlorogenic acid into the standard quinone product

Question 3 (Year 2013)

Q. Recombinant DNA technology (Genetic Engineering) allows genes to be transferred

1. Across different species of plants.
2. From animals to plants.
3. From microorganisms to higher organisms.

Select the correct Solution using the codes given below.

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3**

Answer: D

- Recombinant DNA technology is a major DNA-based tool that opens a new age for modern biotechnology.
- With this technology, a gene or multiple genes can be identified, cut, and inserted into the genome of another organism.
- Golden rice is developed by transferring carrot genes to rice. the new variety of rice contains Vitamin "A".
- Hence Statement 1 correct (across different species of plants)
- In 1953, scientists discovered the structure of DNA, and in 1972, researchers developed a method for cutting and splicing DNA.
- That method became known as recombinant DNA, or rDNA. Since then, researchers have learned how to move genetic material in the form of DNA from one plant or animal to another.
- They can also take DNA from an animal and place it in a plant, and vice versa

Question 4 (Year 2013)

Q. Mycorrhizal Biotechnology has been used in rehabilitating degraded sites because mycorrhiza enables the plants to -

1. Resist drought and increase absorptive area
2. Tolerate extremes of pH
3. Resist disease infestation

Select the correct answer using the codes given below:

- a) 1 only
- b) 1 and 2 only
- c) 1, 2 and 3 only
- d) 2 and 3 only

Answer: C

- Mycorrhizal plants have tolerance to adverse soil, PH, high temp., drought and toxic heavy metals.
- The most important role of Mycorrhizal fungi is to absorb nutrients and water from the soil and transfer them to their hosts.
- Mycorrhizae help plants withstand root diseases.
- Oxalic acid production by mycorrhizal fungi contributes to disease suppression.

Question 5 (Year 2017)

Q. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future?

1. Genome sequencing can be used to identify genetic markers for disease resistance and drought tolerance in various crop plants.
2. This technique helps in reducing the time required to develop new varieties of crop plants.
3. It can be used to decipher the host-pathogen relationships in crops.

Answer: D

Genome sequencing is figuring out the order of DNA nucleotides, or bases, in a genome—the order of As, Cs, Gs, and Ts that make up an organism's DNA.

Genome sequencing of wild varieties of plants can be used to identify disease resistance and drought tolerance genes in various plants and develop new varieties of crop plants in lesser time. Genome sequencing of crop plants can be helpful in deciphering and understanding the host-pathogen relationship in crops.

Therefore, all statements are correct.

Question 6 (Year 2021)

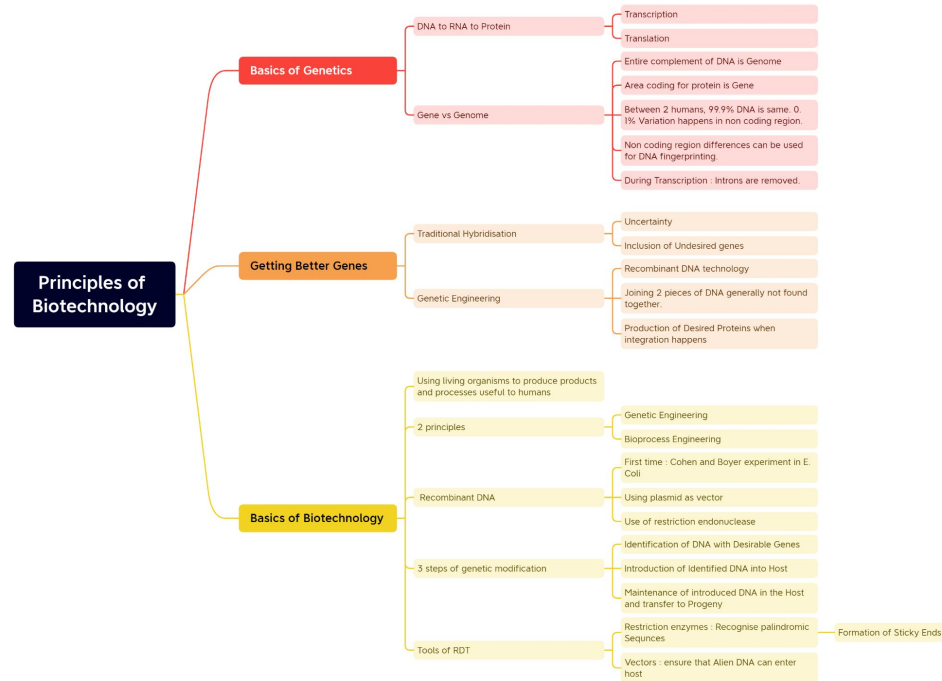
Q. Bollgard I and Bollgard II technologies are mentioned in the context of:

- A. clonal propagation of crop plants
- B developing genetically modified crop plants**
- C. production of plant growth substances
- D. production of biofertilizers

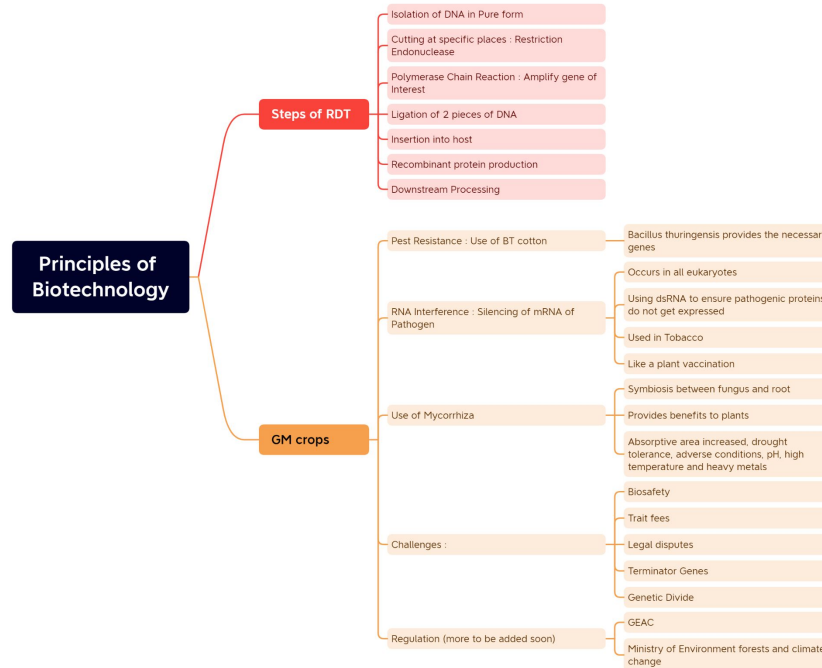
Answer: B

Bollgard I and Bollgard II technologies:

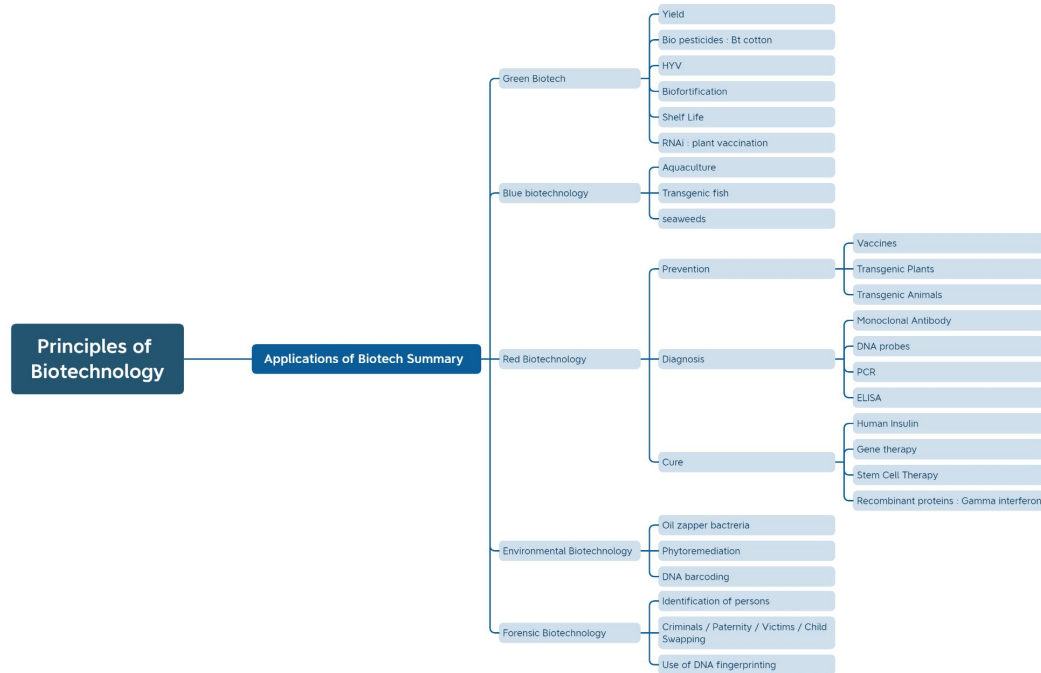
- Bollgard I Bt cotton (single-gene technology) is India's first biotech crop technology approved for commercialization in India in 2002, followed by Bollgard II - double-gene technology in mid-2006, by the Genetic Engineering Approval Committee (GEAC), the Indian regulatory body for biotech crops. Hence, option B is the correct answer.
- Bollgard I cotton provides in-built protection for cotton against destructive American Bollworm *Heliothis Armigera* infestations and contains an insecticidal protein from a naturally occurring soil microorganism, *Bacillus thuringiensis* (Bt).
- Bollgard II technology contains a superior double-gene technology - Cry1Ac and Cry 2Ab which provides protection against bollworms and *Spodoptera* caterpillar, leading to better boll retention, maximum yield, lower pesticides costs, and protection against insect resistance.
- Both, Bollgard II and Bollgard I insect-protected cotton is widely planted around the world as an environmentally friendly way of controlling bollworms.



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